

ANSBACH, Germany

WMO: 107550

Lat: 49.308N

Lon: 10.639E

Elev: 467

StdP: 95.84

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.7	-7.9	-14.2	1.2	-8.8	-10.5	1.6	-5.7	12.8	7.8	11.3	5.4	3.1	40	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	30.1	19.6	28.0	18.7	26.1	18.1	20.5	27.6	19.7	26.1	18.8	24.6	2.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.9	22.0	17.6	13.3	21.5	16.9	12.7	21.0	60.9	27.5	58.2	26.3	55.4	24.7	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 9.6	8.2	7.2	DB	-12.9	32.5	3.1	1.7	-15.1	33.7	-16.9	34.7	-18.6	35.6	-20.8	36.8	(4)
(5)			WB	-12.9	22.3	3.2	1.1	-15.2	23.1	-17.1	23.7	-18.9	24.3	-21.3	25.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	0.1	0.9	5.2	9.9	13.3	16.9	18.9	18.6	14.1	9.6	4.7	1.4		
	DBStd	7.67	4.39	4.56	3.77	4.01	3.87	3.54	3.50	3.41	3.41	3.90	3.70	4.07		
	HDD10.0	1268	306	254	152	50	13	1	0	0	4	57	164	266		
	HDD18.3	3358	564	487	406	252	163	67	37	39	134	272	410	524		
	CDD10.0	1090	0	0	4	48	114	209	275	265	126	44	4	0		
	CDD18.3	138	0	0	0	1	6	26	54	46	6	0	0	0		
	CDH23.3	1236	0	0	0	8	53	234	476	407	57	1	0	0		
	CDH26.7	339	0	0	0	1	6	57	146	122	7	0	0	0		
(14) Wind	WSAvg	3.2	3.8	3.5	3.7	3.1	3.2	2.9	3.0	2.7	2.8	3.0	3.4	3.7		
(15) Precipitation	PrecAvg	684	49	45	48	40	63	71	81	61	54	57	59	57		
	PrecMax	939	110	112	135	108	154	147	157	136	128	191	133	162		
	PrecMin	456	13	9	7	4	16	23	33	18	11	13	0	10		
	PrecStd	106	27	26	29	23	33	30	34	29	26	32	27	32		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	13.7	18.4	24.2	27.5	31.2	33.4	32.8	28.0	22.6	15.4	11.0		
		MCWB	9.5	7.8	10.4	14.1	18.0	21.3	20.3	20.1	17.5	15.1	13.0	9.4		
	2%	DB	9.3	10.1	15.9	21.9	24.9	28.1	30.3	29.9	25.0	19.8	13.2	9.5		
		MCWB	8.1	7.1	9.2	12.7	17.0	19.1	19.3	19.6	16.9	15.0	11.5	8.6		
	5%	DB	7.5	8.4	13.5	19.9	22.5	26.0	28.0	27.5	22.5	17.2	11.7	8.0		
		MCWB	6.4	6.4	8.1	12.0	15.0	18.2	18.5	18.9	15.9	13.8	10.5	7.3		
	10%	DB	6.0	6.9	11.1	17.8	20.5	24.0	25.9	25.2	20.2	15.2	9.9	6.7		
		MCWB	5.1	5.0	7.3	11.3	14.2	17.6	17.9	18.0	15.2	12.9	8.9	6.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	9.2	11.4	15.1	19.3	21.8	21.4	21.6	19.0	16.9	14.0	10.1		
		MCDB	10.8	10.7	16.1	22.6	25.4	28.8	29.6	28.0	24.4	19.7	14.3	10.5		
	2%	WB	8.3	7.7	9.9	13.6	17.7	20.1	20.2	20.5	17.7	15.4	12.1	8.8		
		MCDB	9.2	9.4	14.3	19.7	23.4	26.2	27.3	27.5	23.0	18.1	13.1	9.4		
	5%	WB	6.7	6.6	8.8	12.5	16.3	18.9	19.5	19.6	16.8	14.4	10.5	7.4		
		MCDB	7.5	8.3	12.3	18.4	20.8	24.4	26.0	26.2	21.5	16.6	11.4	8.0		
	10%	WB	5.1	5.1	7.8	11.6	15.0	17.8	18.6	18.7	15.8	13.2	9.0	6.2		
		MCDB	5.7	6.3	10.7	17.0	19.5	22.8	24.5	24.1	19.6	15.0	9.8	6.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	5.8	7.8	9.8	9.4	9.2	9.7	9.8	8.9	7.4	5.0	4.3		
		MCDBR	5.7	8.7	12.4	13.2	12.5	12.8	13.3	13.3	12.4	10.6	7.5	5.9		
		MCWBR	4.8	6.0	6.9	6.5	5.9	5.5	5.1	5.1	5.4	6.3	5.5	5.1		
		MCDBR	5.7	7.9	10.3	12.0	11.1	11.9	11.4	11.9	10.8	9.2	6.9	5.6		
	5% WB	MCWBR	5.1	5.8	6.4	6.4	5.7	5.7	4.8	5.1	5.2	6.0	5.5	5.1		
(40) Clear-Sky Solar Irradiance	taub		0.299	0.322	0.356	0.387	0.395	0.398	0.405	0.399	0.371	0.348	0.322	0.296		
	taud		2.439	2.386	2.333	2.275	2.279	2.299	2.274	2.313	2.372	2.432	2.460	2.439		
	Ebn at Noon		762	824	850	858	864	862	849	837	825	782	723	711		
	Edh at Noon		63	83	104	122	127	126	127	117	99	79	62	55		
(44) All-Sky Solar Radiation	RadAvg		0.87	1.71	2.81	4.26	5.02	5.58	5.29	4.59	3.34	1.99	0.98	0.69		
	RadStd		0.14	0.28	0.36	0.56	0.57	0.45	0.50	0.44	0.43	0.24	0.12	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.40	N/A	N/A	N/A	+20

Nomenclature: See separate page